

Philosophical Logic Essays

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1 Vagueness, Three-Valued Logics

If we adopt a three-valued logic, what grounds are there to prefer Lukasiewicz's truth table for $\rightarrow$ to Kleene's, or vice versa? [2025]

Assess the relative merits of the treatment of the line-drawing Sorites offered by (i) bivalent PL-semantics, (ii) supervaluationism, and (iii) Kleene's three-valued system. [2024]

Which, if any, of the truth tables for $\wedge$ in part (b.ii) give a plausible semantics for conjunction in a vague language? Justify your answer. [2023]

"...supervaluations invalidate our natural mode of deductive thinking" (WILLIAMSON). Discuss. [2022]

Essay Plans

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If we adopt a three-valued logic, what grounds are there to prefer Lukasiewicz's truth table for $\rightarrow$ to Kleene's, or vice versa?

Kleene and Lukasiewicz's truth table for $\rightarrow$ differ in one respect: when $V_{\mathcal{J}}(\phi) = \#$ and $V_{\mathcal{J}}(\psi) = \#$, $\phi \rightarrow \psi$ evaluates as 1 under the Lukasiewicz's scheme while it evaluates as $\#$ under Kleene's truth table. I argue that we ought to accept Kleene's table for the material conditional because it is more coherent with how we think about implication. In particular, Lukasiewicz's treatment of $\rightarrow$ as how much truth is preserved is not congenial with the interpretation of $\#$ as a truth-gap.

Lukasiewicz's truth table differs from Kleene because he interprets the material conditional as how much truth is preserved from the antecedent to the consequent. This is why when ϕ is evaluated at 0, the conditional is automatically evaluated to be true and when ϕ is evaluated at 1 and ψ at 0, there is a loss of

truth which evaluates the conditional $\phi \rightarrow \psi$ at 0. Likewise, when both the antecedent and consequent are evaluated at #, the conditional has preserved all of the truth value from ϕ to ψ , and thus evaluates at 1.

I have two replies to Lukasiewicz. First, I think that Lukasiewicz trades on an erroneous understanding of what # represents. Notice that the Lukasiewicz's truth table is more economically represented if we treat $\# = \frac{1}{2}$ so that

$$LV_{\mathcal{J}}(\phi \rightarrow \psi) = \begin{cases} 1 & \text{if } LV_{\mathcal{J}}(\psi) \geq LV_{\mathcal{J}}(\phi) \\ 1 - (LV_{\mathcal{J}}(\phi) - LV_{\mathcal{J}}(\psi)) & \text{if } LV_{\mathcal{J}}(\psi) < LV_{\mathcal{J}}(\phi) \end{cases}$$

This reveals that Lukasiewicz interprets # as representing a midway between truth and falsity. I do not think that this is true, especially when we consider statements of vagueness. Suppose we have "Tim is bald" as a vague statement – we do not mean that Tim is half way through to being bald but also not enough to be called bald. We mean that there is a gap in being able to assess whether Tim is bald is true. Just because the statement is neither true or false does not imply that it takes a value midway between the two. For if we take the sentence "Tim is vaguely bald" to mean that Tim is literally midway to balding but also not not bald, then it seems that there is no such thing as vagueness. We may not account for the phenomena of vagueness in this way.

Second, it seems to me that Lukasiewicz's interpretation of $\rightarrow$ as truth preservation is also wrong. In standard bivalent semantics, when we think of "If A, then B" as evaluated to be false because when A occurs, B does not occur, it seems to me to be less about truth-preservation than simply the conjunction of the falsity of B and the truth of A. To see what I mean, notice we might express $\rightarrow$ in terms of $\sim (\sim A \vee B)$. If $\rightarrow$ literally functions as truth-preservation, then it should not literally mean $\sim (\sim A \vee B)$ unless we somehow are able to interpret $\sim (\sim A \vee B)$ as a truth-preserving function.

On the flip side, Kleene's truth table seems much more natural – when we evaluate ϕ and ψ as both #, the immediate instinct is to check whether ϕ or ψ being evaluated at # matters, since # is a truth-gap. In the case where both evaluated at #, it clearly matters because ϕ could be true and ψ could be false.

2 Metaphysical Necessity

“There is no interesting link between S5 being the correct modal logic and necessity being truth in all possible worlds.” Assess this claim [2025]

“No plausible deontic logic has the denecessitation property.” Discuss [2024 I]

What, in your view, is the strongest argument against taking S5 to be the correct logic for metaphysical necessity? Expound and critically assess this argument. [2024 II]

“We do not have clear intuitions about the truth values of sentences featuring more than one embedding of modal terms such as “it is necessary that”. This means that there is no reason to prefer S5 over T, or vice versa, as the correct modal logic.’ Discuss. [2023]

“There is no such thing as the real accessibility relation, so there is no such thing as the correct modal logic.” Discuss [2022]

Essay Plans

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“There is no interesting link between S5 being the correct modal logic and necessity being truth in all possible worlds.” Assess this claim.

The quotation suggests that necessity is truth in all possible worlds. In other words, necessity is not merely truth in all possible worlds accessible from some world w , but truth in all possible worlds simpliciter. If we insist on using the notion of accessibility, then we shall say that necessity requires a total or universal relation on the set of worlds. If this is correct, then there is in fact no interesting link between S5 being the correct modal logic and necessity being truth in all possible worlds, because then, S5 is meant to capture some other (weaker) kind of necessity. I argue that there is an interesting link between S5 being the correct modal logic for necessity, because necessity cannot be truth in all possible worlds, but must be more restricted to those accessible from it.

Prima facie, we might take metaphysical necessity to be truth at all possible worlds for quite intuitive reasons. If something is metaphysically necessary, we might reasonably take it to be the strongest kind of necessity possible, which in turn might simply be to be true at every single possible world. This is even, if say, such a possible world is completely isolated from the world in which we inhabit.

The principal reason why I take necessity to not be truth in all possible worlds simpliciter is because we might not be able to express such necessity as a valid modal formula. From part (b.ii), notice that there is no MPL-wff that is valid on a frame $\langle \mathcal{W}, \mathcal{R} \rangle$ iff $\mathcal{R}$ is total on $\mathcal{W}$. Since necessity being truth in all possible worlds just is to take $\mathcal{R}$ being a total or universal relation on $\mathcal{W}$, this means that we have no formula ϕ that can characterise universal relations. In contrast, we have formulae such as $\Box\phi \rightarrow \phi$ that is valid iff $\mathcal{R}$ is reflexive and $\Diamond\Box\phi \rightarrow \Box\phi$. Why might we take this to be important? If we have no MPL formulae that completely characterises totality, then even if we take necessity to mean truth at all possible worlds, we are unable to prove many modal claims which might be valid under a total accessibility relation. In particular, modal logic might fail to be complete if $\mathcal{R}$ is total, which is a huge cost on the semantics and logic of modality.

Contra taking necessity as being truth in all possible worlds, we should then restrict metaphysical necessity as being truth in all accessible possible worlds. If this is true, then S5 is no longer orthogonal to the discussion, but rather central to it. For instance, if we think that the accessibility relation is intransitive on essentialist grounds (per Salmon), then we might reject that S5 is the correct modal logic characterising metaphysical necessity. Alternatively, we might take metaphysical necessity to be the strongest objective

necessity and thus the conjunction of all objective necessities, which entail that the logic of metaphysical necessity must be at least S5 (Williamson). The upshot of this is that, the moment we take accessibility of possible worlds to be relevant to our notion of necessity, then the discussion of what properties our accessibility relation must exhibit is relevant, which implies that there is an interesting link between S5 being the correct modal logic and necessity being truth at all accessible possible worlds.

3 Counterfactuals

"Stalnaker's semantics for would-counterfactuals makes it impossible to give an adequate account of might-counterfactuals." Discuss [2025]

Can Stalnaker's 'limit assumption' be defended against Lewis's criticisms? [2024]

'The only real vice of Stalnaker's semantics is anti-symmetry.' Discuss. [2023]

"Lewis's treatment of 'might' counterfactuals provides a decisive reason to prefer his account of counterfactuals over Stalnaker's. Discuss." [2022]

Essay Plans

Essays

"Stalnaker's semantics for would-counterfactuals makes it impossible to give an adequate account of might-counterfactuals." Discuss [2025]

Stalnaker's semantics for would-counterfactuals is $\phi \Box \rightarrow \psi$ if (i) ϕ is true at no world accessible from i (vacuous case) or (2) ψ is true at the ϕ -world closest to i . I am going to argue that Stalnaker's semantics for would-counterfactuals makes it impossible to give an adequate account of might-counterfactuals because it makes would and might counterfactuals equivalent in the non-vacuous cases in which there are ϕ -worlds.

Stalnaker's semantics for would counterfactuals entails the law of the conditional excluded middle i.e.

$$\models_{SC} (\phi \Box \rightarrow \psi) \vee (\phi \Box \rightarrow \sim \psi)$$

If we were to follow Lewis in defining the might counterfactual as

$$\phi \Diamond \rightarrow \psi =_{df} \sim (\phi \Box \rightarrow \sim \psi)$$

then by the Law of the conditional excluded middle, we yield that $\phi \Diamond \rightarrow \psi$ is equivalent to $(\phi \Box \rightarrow \psi)$ which is an absurd consequence. When we say "If John was free, John would have gone to the party" we mean that in the world in which John was free, John is necessarily at the party. But if we say "If John was free, John might have gone to the party" we mean that it is possible that John is free and yet John is not at the party. This distinction between might and would is entirely lost if we were to adopt Stalnaker's would-counterfactuals.

There are two possible responses to this. First, we might reject Lewis's definition of the might-counterfactual

$$\phi \Diamond \rightarrow \psi =_{df} \sim (\phi \Box \rightarrow \sim \psi)$$

because it might simply beg the question against Stalnaker's semantics. Lewis essentially takes "If John were free, he might have gone to the party" as "It is not the case that (If John were free, he would not have gone to the party)". I think this is quite congenial with our ordinary language intuitions about what "might" means. We typically take might to represent possibility in counterfactual situations. Therefore, to represent the scenario in which it is possible for John to go to the party when he is free, we simply need to eliminate the case in which John necessarily does not go to the party when he is free, which is the negation of the negative would case.

Alternatively, we could examine Stalnaker's proposal of might as $\Diamond(\phi \Box \rightarrow \psi)$. In other words, "If John were free, he might have gone to the party" means "It is possible that (If John were free, he would have gone to the party)". On the face of it, this seems to be immediately defective or at least less convincing than Lewis's alternative. For notice that semantically, Stalnaker's analysis of might means that in some possible world, there is a closest world to that world in which John is free and John goes to the party. But we really want to

say that in the closest possible world where John is free to our world, there is a world in which John goes to the party and John does not go to the party.

The last analysis shows the fundamental defect in Stalnaker's semantics that inhibits it from providing an adequate analysis of might counterfactuals. Because Stalnaker imposes the antisymmetry restriction on his comparative similarity relation, the upshot of this is that Stalnaker cannot represent the intuitive situation of might as the closest possible world in which John is free, and John both goes to the party and John also does not go to the party in a similarly close possible world. Therefore, Stalnaker's semantics for would-counterfactuals makes it impossible to give an adequate account of might counterfactuals.

4 Barcan Formula, Necessary Existents, Necessary Identity

Necessarily, however large a numeral you write down, you could write down a still larger one (e.g. instead of “1000”, you could write “1001”). So the sequence of numerals you could write down is endless: for any member of the sequence, there is a larger one.

Under the most plausible English reading of the argument in part (b), does the conclusion follow logically from the premiss or premisses? Justify your answer, commenting on any difficulties or points of interest. [2025]

‘The correct logic for the strongest objective necessity operator and the unrestricted quantifier validates the Barcan formula and its converse.’ Discuss [2024]

‘Even if one rejects the thesis that existence is non-contingent, there is a clear logical case for the non-contingency of identity.’ Discuss. [2023]

“BF and CBF are metaphysical falsehoods; therefore PC1 is not a logical truth.” Discuss. [2022]

Essay Plans

Necessarily, however large a numeral you write down, you could write down a still larger one (e.g. instead of “1000”, you could write “1001”). So the sequence of numerals you could write down is endless: for any member of the sequence, there is a larger one.

Under the most plausible English reading of the argument in part (b), does the conclusion follow logically from the premiss or premisses? Justify your answer, commenting on any difficulties or points of interest. [2025]

- Claim: the conclusion does follow logically from the premiss.
- Notice that this is valid in SQML but not valid in VDQML – in SQML, we are committed to bare possibilities, so it is always possible that there is a larger numeral but in VDQML, we are not committed to bare possibilities, so it is possible that in our world, we run out of larger numerals
- I think we’re committed to bare possibilities for numerals – there are infinite possible numerals, and all these possible numerals do exist in our world even if we may not be able to write them out now. But also much cheaper ontology – these may not be actually just possible numerals but actual numerals – so we should not express “could write down a still larger one” as possibly a larger one can be written down. Should simply say there is a larger one
- Then it would be valid in both SQML and VDQML – so the above argument being valid in SQML and not in VDQML is actually due to an ambiguity or a mistranslation of taking “could” as possibility when it is a different sense of possibility

“BF and CBF are metaphysical falsehoods; therefore PC1 is not a logical truth.” Discuss. [2022]

- Motivation of the question: BF and CBF are invalid in VDQML axiomatic system that replaces PC1 with PC1’. So, since we want BF and CBF to be invalid in VDQML, then we must replace PC1 with PC1
- Two prongs: are BF and CBF metaphysical falsehoods? and is this a good enough reason to take PC1 as not being a logical truth
- First Prong (BF and CBF are metaphysical falsehoods): agree – intuitively, if it is possible that there exists something that Wittgenstein fathered, then there exists something that Wittgenstein possibly

fathered – but this is false since parentage seems essential to one self. Williamson’s solution of bare possibilities – runs into problems of understanding and cheapening existence. (Hayaki and parsimony of predicates: but I do not think this is the only thing – the real issue is that it cheapens existence as something other than being concrete).

- Second Prong (BF and CBF are to be rejected, and so we should take PC1 as not being a logical truth): We know that $\not\models_{\text{VDQML}} \forall \alpha \phi \rightarrow \phi(\beta/\alpha)$. Why? Because we can have a countermodel in which we have a domain $D = \{u, v\}$ and $D_w = \{u\}$ with $I(F) = \{\langle u, w \rangle\}$. Then clearly, there is a variable assignment g that can assign v to β which renders $F\beta$ false. The intuition here is that the variable assignments can assign things outside the domain of the world.
Is this a good reason to reject PC1? Our intuition for PC1 depends on our assumption that if ϕ is true for any x , then ϕ is true for any y . But this is only if we accept constant domains across worlds. As we have seen we should not accept constant domains across worlds.
We also need to compare this against what we might replace PC1 with – if it is too unintuitive, then it may be less clear if we should reject PC1. One candidate is from (b) where $\forall \beta(\forall \alpha \phi \rightarrow \phi(\beta/\alpha))$ which basically restricts PC1 to only objects in this world. I think this is quite intuitive and restores our understanding of what PC1 is meant to be. In turn, it is not too costly to reject PC1 and PC1’ might be the preferred logical truth.

Essays

‘Even if one rejects the thesis that existence is non-contingent, there is a clear logical case for the non-contingency of identity.’ Discuss. [2023]

The question statement is about necessitism (that everything necessarily exists) and that there is a clear case for the necessity of identity. My argument proceeds in three parts. First, I consider why we might reject the thesis that existence is non-contingent, namely that we have to be committed to bare possibilities. Second, I consider why we should think that identity is non-contingent – that we should support its vindication by SQML. Lastly, I argue that we may not reject necessitism without rejecting the non-contingency of identity, and that this outweighs any ontological costs by bare possibilities.

Necessitism is formulated as $\Box \forall x \Box \exists z x = z$ i.e. it is necessarily the case that anything is necessarily something. On the face of it, this seems obviously false – unicorns do not exist in this world, yet we might take unicorns to be included in the domain of objects. But Williamson justifies the necessity of existence with an ontological commitment to bare possibilities – in short, everything necessarily exists either as something concrete (like you and me) or bare possibilities (i.e. non-concrete existents like unicorns). This ontological commitment is not just costly quantitatively (because it entails many more objects in this world than what we might think) but also qualitatively because bare possibilities are different in kind from concreta. This qualitative profligacy might be extended to understanding the notion of existence. Suppose we want to express the true statement that ‘George W. Bush exists in some worlds, but not in others’. By necessitism, this is false – instead, we have to replace ‘exists’ with ‘is concrete’. But if we do so, then we cannot express that x is essentially F easily. Normally, we would express it as $\Box(\exists x \rightarrow Fx)$. But as we have seen, we must take exists to mean concrete. So suppose we want to say ‘George W. Bush is essentially human’ then we have to say $\Box(\text{George W. Bush is concrete} \rightarrow \text{George W. Bush is human})$. So far so good. But notice that if we want to say that ‘George W. Bush is essentially concrete’ (which is false), we have a vacuously satisfied statement. This implies that we need another kind of predicate to replace $\exists$. But this will not do and we ought to reject necessitism.

Why might there be a clear logical case for the necessity of identity? We have seen that SQML vindicates the validity of the necessity of identity. But notice that so does VDQML because the proof of $\Box \forall x \Box x = x$ relies on very little assumptions. All we have to be committed to is very little – we merely need to be committed to UG, RX and NEC and is compatible with constant domains or variable domains. These, it seems to me, are relatively uncontroversial. There are also metaphysical reasons why we might be convinced by this. If we take names as rigid designators across worlds, then identity holds across worlds. So

there is a clear logical case for the non-contingency identity because it relies on uncontroversial assumptions.

But I want to now argue that there is no way to hold that necessity of identity is true without holding that necessitism is true. Following Williamson, we have the intuitive criteria of the Being Constraint i.e. that having properties requires existence. This is formalised as $\Box\forall x\Box(Fx \rightarrow \exists z x = z)$. But notice that the being constraint and the non-contingency of identity imply $\Box\forall x\Box\exists z x = z$. One option is to reject the Being Constraint – but I think this is too costly for how could a thing be propertied were there no such thing to be propertied? How might this weigh against the qualitative profligacy that Williamson endorses through bare possibilities? I think we have stronger intuitions that the Being Constraint must be true than to think that we should only have one formulaic expression of x is essentially F . Notice that Hayaki’s argument does not entail that we cannot express x is essentially F in some cases, but that we need to (perhaps constantly) think of new predicates to replace existence. But I suspect that for the most part, replacing exists with is concrete will do for many ‘essential’ properties such as being human or having a parent. So, even on this end, it is not extremely costly as Hayaki makes it out to be.

5 Predicate Calculus, Second Order Logic

Essay Plans

‘In second-order logic, = and ι are equally idle.’ Briefly discuss.

- As seen, anything that is expressible by ι (uniqueness such as $Gx\iota xFx$) can be expressed by the formula $\exists x(Gx \wedge \forall y(Fy \rightarrow x = y))$ i.e. there is exactly one F that is a G .

- SOL can express identity by

$$x = y \iff_{\text{def}} \forall F(Fx \leftrightarrow Gx)$$

- So, if SOL can express identity, and identity can express ι , then SOL can also fully eliminate ι
- Prima facie equally idle
- Objection: = cannot be eliminated in SOL as above because it relies on assuming Leibniz’s indiscernibility of identicals which is a substantive metaphysical thesis, but we have reason to reject this (for instance, we a lump of clay and a statue may be qualitatively identical but not actually identical). One reason for this is because identity is necessary. We can move from $x = x$ to $\Box x = x$. But it is clear that the clay and the statue are not necessarily identical.

Essay

“Anything expressible in the language of second-order logic can be equally well-expressed in the language of first order set theory” Assess [2025]

Second Order Logic takes predicates to be quantifiable over while the language of first order set theory takes $\in$ as part of its primitive vocabulary. In particular, we need not actually take $\in$ to be part of its primitive vocabulary and can simply use a predicate Exy to express this instead. If anything expressible in SOL can be equally well-expressed in the language of first-order set theory, then this is reminiscent of Quine’s objection that SOL is ‘set-theory is sheep’s clothing’. I am going to argue that it is not true that anything expressible in SOL can be equally well-expressed in the language of first-order set theory, and that it can express more validities.

From parts (a) and (b) of the question, notice that P1 and I1 are not PC validities, but that P2 and I2 (their SOL counterparts) are SOL-valid. What this means is that it is clear that even elementary truths like any two objects form a set cannot be equally expressed (i.e. not true in all and only the same models) in first order set theory as SOL. Furthermore, not even weaker claims P1 or I1 is valid under FOL. The upshot of this is that SOL can take what first order set theory take as basic premises (such as the axiom of pairing) and make them valid. As suggested earlier, this may be reminiscent of Quine’s scepticism against SOL – namely that

it smuggles in truths of set theory and because set theory is not logic, this means that second-order logic is not actually logic.

However, we may think that even though sentences in SOL and first-order set theory may mean the same thing in idiomatic English, they do not actually mean the same thing. Following Boolos, he argues that no valid second-order sentence can assert the same thing as any theorem of set theory. The basic reason is that second order sentences asserts something only with respect to an interpretation, whose domain may not be taken to contain all sets. If, for contradiction, the sentence were to assert what any particular set theoretic statement asserts, its domain would have to contain all sets. For instance $\exists X \forall x Xx$ (there is a set of all sets) is valid, but does not assert there is a universal set which according to ZFC, is false. Instead, it asserts that there is a subset of the domain to which everything in the domain belongs.

Overall, second order statements do not express the same thing as first order set theory because their domain may not be taken to contain all sets.

‘One respect in which identity theory seems a nearer neighbour to logic than to mathematics is its completeness’ (QUINE). Is the incompleteness of second-order logic a good reason to think that it is not really logic, but mathematics? [2020]

The incompleteness of second-order logic is the result that not every valid formula in second order logic can be proven. This results from the fact that second-order logic is not compact as seen in part (b). According to Gödel, any recursively axiomatisable system T strong enough to encode arithmetic is incomplete in this sense, but first order logic (what Quine calls identity theory) is complete. This essay argues that the incompleteness of second-order logic is a good reason to think that it is not really logic, but mathematics in three steps. First, I argue that logic should be taken to be compact and complete because logic should focus solely on the formal. Second, I consider other reasons why SOL is not really logic, namely ontological commitments and treating predicates name positions as contend that incompleteness is a better reason. Lastly, I consider whether restoring completeness through Henkin semantics can defuse the objection from incompleteness, concluding that it does not.

Whether or not second order logic is considered logic or mathematics depends on what we define logic to be. One reasonable definition that both Quine and Boolos seem to agree on (but differ on what this entails) is that logic should be neutral on theory. This means that it should not presuppose ontological commitments or favour some other substantive theory. However, it seems difficult to parse whether incompleteness makes logic no longer neutral on theory – it seems either way, whether SOL is complete (implying that all validities must be provable) or incomplete seems to make some substantive commitment. So, Quine and Boolos’s ideal of theory-neutrality is not enough. Here, I think one additional requirement for what logic should be is following Gödel: namely that logic is a solely formal enterprise. This means that deductive systems (such as proof systems) should be checkable that they are correct. What this means is that we need all correct proofs to be finite in nature (otherwise, we cannot check such proofs), which in turn implies that the logic must be complete and compact. Mathematics, on the other hand, need not be a solely formal enterprise, which gives it the leeway to be incomplete. This seems to suggest that incompleteness is a good reason to think that SOL is nearer to mathematics.

We may think that incompleteness may not be an important reason to think why SOL is not logic. After all, it seems more intuitive that the most important requirement for logic is theory-neutrality. So, the objection from incompleteness is good to the extent that objections from name positions and ontological commitments do not succeed. Quine argues that SOL is not logic because SOL embeds set theory into its conception of ‘logic’ and Set theory is not logic. Quine draws an analogy between ordinary quantification in FOL and name positions – when we say $\forall x(x \text{ walks})$, we treat x as a name position such as ‘Boolos walks’ or ‘Matthew walks’. Likewise, according to Quine, when we quantify over predicates, we treat predicates as name positions as well for some sort of entity. Because of this, logicians treat predicate letters like sets (if we identify properties with sets). If we treat predicate variables as name positions for sets, then we smuggle in truths of set theory such as $\exists G \forall x (Gx \leftrightarrow Fx)$ as logical validities. This makes a substantive commitment to set theory,

which is contra the mission of making logic neutral. But I do not think this is a successful objection. For one, it may be amiss to think that just because we treat the predicate F as having a range, means that we treat it as a name position. Furthermore, even if it is true that SOL smuggles in truths of set theory, we can defuse the objection in two ways. First, as we can see from above, SOL smuggles in a very limited range of set theoretical validities. While it makes the formula $\exists X \exists Y \sim \forall x (Xx \leftrightarrow Yx)$ i.e. there are at least two sets valid, it does not even make the existence of a two-membered set valid. Second, following Boolos, we may think that set theory is distinctively logical. This is because anything can belong to a set or be related to something. In this way, Quine's objections lose force and the argument from incompleteness seems like the best objection.

Lastly, I want to consider the possibility of restoring completeness in SOL via Henkin semantics. While I do not go into the technical details of Henkin semantics, what is important is that Henkin semantics force second order quantifiers to only range over a specified collection of relations/functions (and not all possible ones). However, this makes SOL lose much of what makes it compelling – it loses much of its expressive power such as expressing categoricity. Furthermore, if we have to make some arbitrary restrictions on what second-order quantifiers can range over, this seems to be itself a substantive, non-neutral commitment. It is my view that the cost of restoring completeness in SOL renders it unconvincing over FOL, while also not rendering it topic neutral as desired.